Tuple

A tuple in Python is a collection used to store multiple items in a single variable.

Main features of a tuple

1. Ordered: Items maintain their order.

2. Immutable: You cannot change, add, or remove items after creation.

3. Allows duplicates: (10, 20, 10) is valid.

4. Different data types: (10, "Reeta", 5.5) is valid.

5. Indexing: You can access items using their index.

1.Creating a Tuple

In [5]:
```
a=(1,2,3,4,5)
print(a)
```

```
(1, 2, 3, 4, 5)
```

In [7]:
```
a=(1,2,3,4,5)
print(type(a))
```

```
<class 'tuple'>
```

In [9]:
```
a=("Reeta","Mano")
print(type(a))
```

```
<class 'tuple'>
```

In [11]:
```
a="Reeta","Elayaraja"
print(type(a))
```

```
<class 'tuple'>
```

In [13]:
```
a=1,2,3,4,5
print(type(a))
```

```
<class 'tuple'>
```

2.Access Tuple

In [17]:
```
a=(3,2,1,6,7,9,5)
print(a[2])
```

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In [19]:

```
a=(3,2,1,6,7,9,5)
print(a[-1])
```

5

In [21]:

```
a=(3,2,1,6,7,9,5)
print(a[2:6])
```

(1, 6, 7, 9)

In [27]:

```
a = (3, 2, 1, 6, 7, 9, 5)
print(a[-5:-2])
```

(1, 6, 7)

3.Check if Item Exists

In [30]:

```
a=(1,2,3,5,7)
if 4 in a:
    print("yes")
else:
    print("no")
```

no

4.update typle

In [33]:

```
a=(10,12,30,40,50)
b=list(a)
b[1]=20
a=tuple(b)
print(a)
```

(10, 20, 30, 40, 50)

5.add item

In [38]:

```
a=(10,20,30,40)
b=list(a)
b.append(50)
a=tuple(b)
print(a)
```

(10, 20, 30, 40, 50)

In [3]:

```
a=(10,20,30,40)
y=(50,)
a +=y
print(a)
```

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```
(10, 20, 30, 40, 50)
```

6.remove item

In [45]:

```
a=(10,20,30,40)
b=list(a)
b.remove(30)
a=tuple(b)
print(a)
```

```
(10, 20, 40)
```

7.del

In [48]:

```
a=(10,20,30,40)
del a
print(a)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[48], line 3
      1 a=(10,20,30,40)
      2 del a
----> 3 print(a)

NameError: name 'a' is not defined
```

8.unpack typle

In [53]:

```
a=(10,20,30,40)
(A,B,C,D)=a
print(A)
print(B)
print(C)
print(D)
```

```
10
20
30
40
```

In [55]:

```
a=(10,20,30,40)
(A,B,*C)=a
print(A)
print(B)
print(C)
```

```
10
20
[30, 40]
```

In [57]:

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```
a=(10,20,30,40)
(A,*B,C)=a
print(A)
print(B)
print(C)
```

```
10
[20, 30]
40
```

9.count

In [60]:

```
a=(1,2,3,5,6,1,2,36,1,1)
b=a.count(1)
print(b)
```

```
4
```

10.index

In [63]:

```
a=(10,20,30,40)
b=a.index(20)
print(b)
```

```
1
```

Practie Question

In [ ]:

```
1. Write a Python program to find the length of a tuple.

2. Write a Python program to find the maximum and minimum values in a tuple.

3. Write a Python program to count how many times a given element appears in a tuple.

4. Write a Python program to reverse a tuple using slicing.

5. Write a Python program to find the index of a given element in a tuple.
```

1. Write a Python program to find the length of a tuple.

In [67]:

```
a=(21,56,43,23,12,78,65)
print(len(a))
```

```
7
```

2. Write a Python program to find the maximum and minimum values in a tuple.

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In [70]:

```
a=(1,2,3,4,5)
b=max(a)
c=min(a)
print(b)
print(c)
```

```
5
1
```

3. Write a Python program to count how many times a given element appears in a tuple.

In [73]:

```
a=(1,2,4,6,7,8,2,3,4,2,6,8,4,2,2,1)
b=a.count(2)
print(b)
```

```
5
```

4. Write a Python program to reverse a tuple using slicing.

In [78]:

```
a=(1,2,3,4,5,6,7,8,9)
print(a[::-1])
```

```
(9, 8, 7, 6, 5, 4, 3, 2, 1)
```

5. Write a Python program to find the index of a given element in a tuple.

In [81]:

```
a=(1,2,3,4,5,6,7,8,9)
b=a.index(3)
print(b)
```

```
2
```

6. Write a Python program to concatenate two tuples.

In [84]:

```
a=(1,2,3)
b=(4,5,6)
print(a+b)
```

```
(1, 2, 3, 4, 5, 6)
```

7.Write a program to change the second element of a tuple to 100 without directly modifying the tuple.

In [87]:

```
a=(1,2,3,4,5)
b=list(a)
b[2] = 100
a=tuple(b)
print(a)
```

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```
(1, 2, 100, 4, 5)
```

8.Write a program to unpack (10, 20, 30) into three variables and print them.

In [90]:

```
a=(10,20,30)
(A,B,C)=a
print(A)
print(B)
print(C)
```

```
10
20
30
```

9.Write a Python program to add the element 40 to the end of a tuple (10, 20, 30) without using loops or conditions.

In [93]:

```
a=(10,20,30)
b=list(a)
b.append(40)
a=tuple(b)
print(a)
```

```
(10, 20, 30, 40)
```

10.Write a Python program to remove the element 20 from the tuple (10, 20, 30, 40) without using loops or conditions.

In [96]:

```
a=(10,20,30,40)
b=list(a)
b.remove(20)
a=tuple(b)
print(a)
```

```
(10, 30, 40)
```

In [ ]:

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